

Tae Jun Ham

Senior Staff Software Engineer & Tech Lead @ Google

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About

I am a **Senior Staff Software Engineer and Tech Lead at Google**, focusing on **hardware/software performance, resource efficiency, and co-design** for Google's datacenter fleet and distributed systems infrastructure. Specializing in **cross-stack optimization**, my work bridges distributed systems and silicon microarchitecture to untangle complex system behaviors, optimize foundational infrastructure, and guide future SoC designs.

Prior to joining Google, I was a postdoctoral researcher at Seoul National University. I received my **Master's and Ph.D. in Electrical Engineering from Princeton University** under the supervision of Professor Margaret Martonosi, and a B.S.E. degree from Duke University. My research centers on hardware-software co-design for emerging applications and data access optimizations across systems and architectures.

Throughout my career, I have published **30+ works** in top computer architecture and systems venues (ISCA, ASPLOS, MICRO, HPCA, ATC, MLSys, etc.). My research emphasizes architectural acceleration and co-design, with key highly-cited contributions including early Transformer/Attention accelerators and a graph analytics accelerator. My work has been recognized with the **Best Paper Award at MICRO-49**, **IEEE Micro Top Picks** (including Honorable Mention), a **Best Paper Award Nomination at ISPASS**, and the **Samsung Scholarship**. I also actively serve on program committees for leading conferences in the field.

Education

Princeton University, Princeton, NJ USA

Sep 2012 – Jun 2018

School of Engineering and Applied Science

- **Ph.D and M.A** in Electrical Engineering
- **Dissertation:** Data Access Optimization in Accelerator-oriented Heterogeneous Architecture through Decoupling and Memory Hierarchy Specialization
- **Advisor:** Professor Margaret Martonosi and Professor Juan Luis Aragon

Duke University, Durham, North Carolina USA

Aug 2009 – Dec 2011

Pratt School of Engineering

- **B.S.E** in Electrical and Computer Engineering (GPA : 3.95/4.00)
- *Summa Cum Laude, with Distinction in Electrical and Computer Engineering*

Experience

Google, Sunnyvale, California, USA

Sep 2021 – Present

Senior Staff Software Engineer & Tech Lead

Lead technical strategy for cross-stack performance, hardware/software co-design, and resource efficiency across Google's datacenter fleet and distributed systems. Previously was a tech lead manager for a ~20-person team before transitioning to a strategic technical lead role.

- **Datacenter Efficiency:** Led fleet-wide optimization initiatives across compute, memory, and storage, delivering significant capacity savings.
- **Hardware/Software Co-Design:** Bridge distributed systems and silicon microarchitecture to evaluate, optimize, and guide the architectural design of future SoC and server platforms.
- **ML for Systems:** Focus on leveraging machine learning to drive system efficiency and performance, and unlock new opportunities in hardware/software co-design.

Seoul National University, Seoul, Republic of Korea

Jul 2018 – Jul 2021

Postdoctoral Researcher

Focused on Computer Architecture and Systems research.

- **Supervisor:** Jae W. Lee.

- (Note: My position at Seoul National University also fulfilled my mandatory military service duty)

Microsoft Research, Cambridge, UK

May – Aug 2016

Graduate Research Intern

Investigated and designed an efficient secure memory architecture featuring near-data computation.

- **Collaborator:** Stavros Volos.

Intel Labs — Parallel Computing Lab, Santa Clara, USA

May – Nov 2015

Graduate Technical Intern

Conducted research and development on a custom hardware accelerator tailored for graph analytics.

- **Collaborator:** Lisa Wu Wills.

AMD Research, Austin, USA

Jun – Aug 2013

Co-op Engineer

Explored heterogeneous memory systems, focusing specifically on stacked DRAM architectures and performance optimization.

- **Collaborator:** Joseph L. Greathouse.

Samsung Advanced Institute of Technology, Yongin, Republic of Korea

Jun – Aug 2012

Research Intern

Investigated GPU branch divergence problems and proposed architectural solutions to improve execution efficiency.

- **Collaborator:** Yeon-Gon Cho.

Duke University — Systems Architecture Integration Lab, Durham, USA

Jan – May 2012

Research Assistant

Research on efficient control and management of the heterogeneous memory system.

- **Advisor:** Benjamin C. Lee.

Honors and Awards

- **IEEE MICRO Top Picks (2021):** Genesis paper selected as one of top 12 computer architecture papers of 2020
- **IEEE MICRO Top Picks Honorable Mention (2021):** Graphene paper selected as one of top 24 computer architecture papers of 2020
- **ISPASS Best Paper Award Nominee (2020):** MosaicSim paper
- **MICRO-49 Best Paper Award (2016):** Graphicionado paper
- **IEEE MICRO Top Picks Honorable Mention (2016):** DeSC paper (Top 23 computer architecture papers of 2015)
- **Facebook Graduate Fellowship Finalist (2016-2017),** Facebook, Inc.
- **Gordon Y.S. Wu Fellowship (2012-2017),** Princeton University
- **Samsung Scholarship (2012-2017),** Up to \$50,000 per year for five years of graduate studies
- **Summa Cum Laude (2011),** Duke University

Selected Publications

[EMNLP '25] Reliable and Cost-Effective Exploratory Data Analysis via Graph-Guided RAG

Mossad Helali, Yutai Luo, **Tae Jun Ham**, Jim Plotts, Ashwin Chaugule, Jichuan Chang, Parthasarathy Ranganathan, Essam Mansour

The 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)

[SAC '25] SkipLSM: Fast Retrieval of Hot Key-Value Pairs on LSM Tree

Jongsung Lee, Sam Son, Jonghyun Bae, Yunho Jin, **Tae Jun Ham**, Jae W. Lee

The 40th ACM/SIGAPP Symposium on Applied Computing (SAC)

Acceptance Rate : TBD

[ACM TOS '24] An LSM tree augmented with b+ tree on nonvolatile memory

Donguk Kim, Jongsung Lee, Keun Soo Lim, Jun Heo, **Tae Jun Ham**, Jae W. Lee

ACM Transactions on Storage (TOS)

[ISCA@50 Retrospective] Genesis: A Hardware Acceleration Framework for Genomic Data Analysis

Lisa Wu Wills, **Tae Jun Ham**, Jae W. Lee, Krste Asanovic

The 50th Annual IEEE/ACM International Symposium on Computer Architecture (Retrospective)

98 papers were selected out of 1000+ papers across 25-years of ISCA history

[HPCA '22] Mithril: Cooperative Row Hammer Protection on Commodity DRAM Leveraging Managed Refresh

Michael Jaemin Kim, Jaehyun Park, Yeonhong Park, Wanju Doh, Namhoon Kim, **Tae Jun Ham**, Jae W. Lee, Jung Ho Ahn

The 28th IEEE International Symposium on High-Performance Computer Architecture (HPCA)

Acceptance Rate : 80/262 $\approx$ 30%

[ACM TECS '22] MaPHeA: A framework for lightweight memory hierarchy-aware profile-guided heap allocation

Deok-Jae Oh, Yaebin Moon, Do Kyu Ham, **Tae Jun Ham**, Yongjun Park, Jae W. Lee, Jung Ho Ahn, Eojin Lee

ACM Transactions on Embedded Computing Systems (TECS)

[MLSys '22] ULPPACK: Fast Sub-8-bit Matrix Multiply on Commodity SIMD Hardware

Jaeyeon Won, Jeyeon Si, Sam Son, **Tae Jun Ham**, Jae W. Lee

Proceedings of Machine Learning and Systems (MLSys)

[HPCA '22] ANNA: Specialized Architecture for Approximate Nearest Neighbor Search

Yejin Lee, Hyunji Choi, Sunhong Min, Hyunseung Lee, Sangwon Baek, Dawoon Jeong, Jae W. Lee, **Tae Jun Ham**

The 28th IEEE International Symposium on High-Performance Computer Architecture (HPCA)

Acceptance Rate : 80/262 $\approx$ 30%

[ECCV '22] L3: Accelerator-Friendly Lossless Image Format for High-Resolution, High-Throughput DNN Training

Jonghyun Bae, Woohyeon Baek, **Tae Jun Ham**, Jae W. Lee

European Conference on Computer Vision (ECCV)

Acceptance Rate : 1650/5803 $\approx$ 28%

[IEEE TC '22] Architecting a Flash-based Storage System for Low-cost Inference of Extreme-scale DNNs

Yunho Jin, Shine Kim, **Tae Jun Ham**, Jae W. Lee

IEEE Transactions on Computers (TC)

[ASPLOS '21] MERCI: Efficient Embedding Reduction on Commodity Hardware via Sub-Query Memoization

Yejin Lee, Seong Hoon Seo, Hyunji Choi, Hyoung Uk Sul, Soosung Kim, Jae W. Lee, **Tae Jun Ham**

The 26th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)

Acceptance Rate : 75/398 $\approx$ 19%

[FAST '21] FlashNeuron: SSD-Enabled Large-Batch Training of Very Deep Neural Networks

Jonghyun Bae, Jongsung Lee, Yunho Jin, Sam Son, Shine Kim, Hakbeom Jang, **Tae Jun Ham**, Jae W. Lee

USENIX Conference on File and Storage Technologies (FAST)

Acceptance Rate : 28/130 $\approx$ 21%

[FAST '21] Behemoth: A Flash-centric Training Accelerator for Extreme-scale DNNs

Shine Kim, Yunho Jin, Gina Sohn, Jonghyun Bae, **Tae Jun Ham**, Jae W. Lee

USENIX Conference on File and Storage Technologies (FAST)

Acceptance Rate : 28/130 $\approx$ 21%

[HPCA '21] Layerweaver: Maximizing Resource Utilization of Neural Processing Units via Layer-Wise Scheduling

Young H. Oh, Seonghak Kim, Yunho Jin, Sam Son, Jonghyun Bae, Jongsung Lee, Yeonhong Park, Dong Uk Kim, **Tae Jun Ham**, Jae W. Lee

The 27th IEEE International Symposium on High-Performance Computer Architecture (HPCA)

Acceptance Rate : 63/258 $\approx$ 24%

[IEEE Micro Top Picks '21] Accelerating Genomic Data Analytics with Composable Hardware Acceleration Framework

Tae Jun Ham, David Bruns-Smith, Brendan Sweeney, Yejin Lee, Seong Hoon Seo, U Gyeong Song, Young H. Oh, Krste Asanovic, Jae W. Lee, Lisa Wu Wills

IEEE Micro Special Issue on Top Picks from the 2020 Computer Architecture Conferences

[ISCA '21] ELSA: Hardware-Software Co-design for Efficient, Lightweight Self-Attention Mechanism in Neural Networks

Tae Jun Ham, Yejin Lee, Seong Hoon Seo, Soosung Kim, Hyunji Choi, Sung Jun Jung, Jae W. Lee
The 48th ACM/IEEE International Symposium on Computer Architecture (ISCA)
Acceptance Rate : 76/406 $\approx$ 19%

[ISCA '21] BOSS: Bandwidth-Optimized Search Accelerator for Storage-Class Memory

Jun Heo, Seung Yul Lee, Sunhong Min, Yeonhong Park, Sung Jun Jung, **Tae Jun Ham**, Jae W. Lee
The 48th ACM/IEEE International Symposium on Computer Architecture (ISCA)
Acceptance Rate : 76/406 $\approx$ 19%

[ATC '21] ASAP: Fast Mobile Application Switch via Adaptive Prepaging

Sam Son, Seung Yul Lee, Yunho Jin, Jonghyun Bae, Jinkyu Jeong, **Tae Jun Ham**, Jae W. Lee, Hongil Yoon
USENIX Annual Technical Conference (ATC)
Acceptance Rate : 64/341 $\approx$ 19%

[LCTES '21] MaPHeA: A Lightweight Memory Hierarchy-aware Profile-guided Heap Allocation

Deok-Jae Oh, Yaebin Moon, Eojin Lee, **Tae Jun Ham**, Jae W. Lee, Jung Ho Ahn
ACM SIGPLAN/SIGBED Conference on Languages, Compilers, Tools and Theory of Embedded Systems (LCTES)
Acceptance Rate : 15/37 $\approx$ 40%

[ICCAD '20] Unlocking Wordline-Level Parallelism for Fast Inference on RRAM-Based DNN Accelerator

Yeonhong Park, Seung Yul Lee, Hoon Shin, Jun Heo, **Tae Jun Ham**, Jae W. Lee
The 39th IEEE/ACM International Conference on Computer-Aided Design (ICCAD)
Acceptance Rate : 127/470 $\approx$ 27%

[MICRO '20] Graphene: Strong yet Lightweight Row Hammer Protection

Yeonhong Park, Woosuk Kwon, Eojin Lee, **Tae Jun Ham**, Jung Ho Ahn, Jae W. Lee
The 53rd IEEE/ACM International Symposium on Microarchitecture (MICRO)
Acceptance Rate : 82/424 $\approx$ 19% (● **IEEE Micro Top Picks Honorable Mention**)

[ISCA '20] Genesis: A Hardware Acceleration Framework for Genomic Data Analysis

Tae Jun Ham, David Bruns-Smith, Brendan Sweeney, Yejin Lee, Seong Hoon Seo, U Gyeong Song, Young H. Oh, Krste Asanovic, Jae W. Lee, Lisa Wu
The 47th ACM/IEEE International Symposium on Computer Architecture (ISCA)
Acceptance Rate : 77/428 $\approx$ 18% (● **IEEE Micro Top Picks**)

[ISCA '20] A Case for Hardware-Based Demand Paging

Gyusun Lee*, Wenjing Jin*, Wonsuk Song, Jeonghun Gong, Jonghyun Bae, **Tae Jun Ham**, Jae W. Lee, Jinkyu Jeong
The 47th ACM/IEEE International Symposium on Computer Architecture (ISCA)
Acceptance Rate : 77/428 $\approx$ 18% (*Two authors contributed equally.)

[ISCA '20] A Specialized Architecture for Object Serialization with Applications to Big Data Analytics

Jaeyoung Jang, Sung Jun Jung, Sunmin Jeong, Jun Heo, Hoon Shin, **Tae Jun Ham**, Jae W. Lee
The 47th ACM/IEEE International Symposium on Computer Architecture (ISCA)
Acceptance Rate : 77/428 $\approx$ 18%

[ISPASS '20] MosaicSim: A Lightweight, Modular Simulator for Heterogeneous Systems

Opeoluwa Matthews, Aninda Manocha, Davide Giri, Marcelo Orenes-Vera, Esin Tureci, Tyler Sorensen, **Tae Jun Ham**, Juan Luis Aragon, Luca P. Carloni, Margaret Martonosi
IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS)
Acceptance Rate : 25/73 $\approx$ 34% (● **Nominated for the Best Paper Award**)

[ASPLOS '20] IIU: Specialized Architecture for Inverted Index Search

Jun Heo, Jaeyeon Won, Yejin Lee, Shivam Bharuka, Jaeyoung Jang, **Tae Jun Ham**, Jae W. Lee
The 25th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)
Acceptance Rate : 86/476 $\approx$ 18%

[HPCA '20] A³: Accelerating Attention Mechanisms in Neural Networks with Approximation

Tae Jun Ham, Sung Jun Jung, Seonghak Kim, Young H. Oh, Yeonhong Park, Yoonho Song, Jung-Hun Park, Sanghee Lee, Kyoung Park, Jae W. Lee, Deog-Kyoon Jeong
The 26th IEEE International Symposium on High Performance Computer Architecture (HPCA)
Acceptance Rate : 48/235 $\approx$ 20%

[MICRO '19] Charon: Specialized Near-Memory Processing Architecture for Clearing Dead Objects in Memory
Jaeyoung Jang, Jun Heo, Yejin Lee, Jaeyeon Won, Seonghak Kim, Sung Jun Jung, Hakbeom Jang, **Tae Jun Ham**, Jae W. Lee

The 52nd IEEE/ACM International Symposium on Microarchitecture (MICRO)

Acceptance Rate : 79/343 $\approx$ 23%

[IEEE Micro '19] SSDStreamer: Specializing I/O Stack for Large-Scale Machine Learning

Jonghyun Bae, Hakbeom Jang, Jeonghun Gong, Wenjing Jin, Shine Kim, Jaeyoung Jang, **Tae Jun Ham**, Jinkyu Jeong, Jae W. Lee

IEEE Micro, September 2019

[ATC '19] Asynchronous I/O Stack: A Low-latency Kernel I/O Stack for Ultra-Low Latency SSDs

Gyusun Lee, Seokha Shin, Wonsuk Song, **Tae Jun Ham**, Jae W. Lee, Jinkyu Jeong

USENIX Annual Technical Conference (ATC)

Acceptance Rate : 71/356 $\approx$ 20%

[ATC '19] Practical Erase Suspension for Modern Low-latency SSDs

Shine Kim, Jonghyun Bae, Hakbeom Jang, Wenjing Jin, Jeonghun Gong, Seungyeon Lee, **Tae Jun Ham**, Jae W. Lee

USENIX Annual Technical Conference (ATC)

Acceptance Rate : 71/356 $\approx$ 20%

[ACM TACO '19] Efficient Data Supply for Parallel Heterogeneous Architectures

Tae Jun Ham, Juan L Aragon, Margaret Martonosi

ACM Transactions on Architecture and Code Optimization (TACO)

Presented on **HiPEAC** 2020 Conference

[ACM TACO '17] Decoupling Data Supply from Computation for Latency-Tolerant Communication in Heterogeneous Architectures

Tae Jun Ham, Juan L Aragon, Margaret Martonosi

ACM Transactions on Architecture and Code Optimization (TACO)

[MICRO '16] Graphicionado: A High-Performance and Energy-Efficient Accelerator for Graph Analytics

Tae Jun Ham, Lisa Wu, Narayanan Sundaram, Nadathur Satish, Margaret Martonosi

The 49th IEEE/ACM International Symposium on Microarchitecture (MICRO)

Acceptance Rate : 61/283 $\approx$ 22% (● **MICRO-49 Best Paper Award**)

[MICRO '15] DeSC: Decoupled Supply-Compute Communication Management for Heterogeneous Architectures

Tae Jun Ham, Juan L Aragon, Margaret Martonosi

The 48th IEEE/ACM International Symposium on Microarchitecture (MICRO)

Acceptance Rate : 61/283 $\approx$ 22% (● **IEEE Micro's Top Picks Honorable Mention**)

[HPCA '13] Disintegrated Control for Energy-Efficient and Heterogeneous Memory Systems

Tae Jun Ham, Bharath K. Chelepalli, Neng Xue, Benjamin C. Lee

The 19th IEEE International Symposium on High Performance Computer Architecture (HPCA)

Acceptance Rate : 51/249 $\approx$ 20%

Patents

- **Apparatus and method with scheduling** (US Patent 12,524,267)
with Jae Wook Lee, Younghwan Oh, and Yunho Jin
- **Scheduler, method of operating the same, and accelerator apparatus including the same** (US Patent 12,277,440)
with Seung Wook Lee, Jae Wook Lee, Young Hwan Oh, and Seng Hak Kim
- **Processor, method of operating the processor, and electronic device including the same** (US Patent 12,327,179)
with Seung Wook Lee, Jaeyeon Won, and Jae Wook Lee
- **Layer-wise scheduling on models based on idle times** (US Patent 12,099,869)
with Seung Wook Lee, Younghwan Oh, Jaewook Lee, Sam Son, and Yunho Jin
- **Hardware accelerator performing search using inverted index structure and search system including the hardware accelerator** (US Patent 11,544,270)

- with Jun Heo, Jaeyeon Won, Yejin Lee, Jaeyoung Jang, and Jae Wook Lee
- **Hammer refresh row address detector, and semiconductor memory device and memory module including the same** (US Patent 11,568,917)
with Hoon Shin, Yeonhong Park, Jaewook Lee, Eojin Lee, Woosuk Kwon, and Jungho Ahn
- **Method for candidate selection and accelerator for performing candidate selection** (US Patent 11,636,173)
with Seonghak Kim, Sungjun Jung, Younghwan Oh, Jaewook Lee, Deog-Kyoon Jeong, and Minsoo Lim
- **Device for accelerating self-attention operation in neural networks** (US Patent App. 17/864,235)
with Ye Jin Lee, Seong Hoon Seo, Soo Sung Kim, Hyun Ji Choi, Jae W. Lee, and Sung Jun Jung
- **Accelerator system for training deep neural network model using nand flash memory and operating method thereof** (US Patent App. 18/089,141)
with Jae W. Lee, Yunho Jin, Jong Hyun Bae, and Gin A Sohn
- **Method for processing page fault by processor** (US Patent 11,436,150)
with Jinkyu Jeong, Jae Wook Lee, and Wenjing Jin
- **Electronic device and method with scheduling** (US Patent App. 17/195,748)
with Seung Wook Lee, Younghwan Oh, Jaewook Lee, Sam Son, and Yunho Jin
- **Instruction, Circuits, and Logic for Graph Analytics Acceleration** (US Patent App. 15/089,232)
with Lisa K. Wu, Narayanan Rajagopalan Satish, and Narayanan Sundaram

Professional Service

- **MICRO**: Program Committee (2026), External Review Committee (2025, 2024), Student Research Competition Selection Committee (2023)
- **HPCA**: Program Committee (2025, 2026), Industry Track Program Committee (2026)
- **ISCA**: Program Committee (2025), External Review Committee (2024)
- **ASPLOS**: Program Committee (2023, 2024), External Review Committee (2019)
- **IISWC**: Program Committee (2024)
- **YArch**: Program Committee (2024)
- **IEEE Micro Top Picks**: Program Committee (2023)
- **CGO**: Web Chair (2021, 2022)

Paper Reviews

- IEEE Transactions on Computers (TC), IEEE TVLSI, IEEE TMC, IEEE TCAD
- IEEE Computer Architecture Letters (CAL), IEEE Micro
- ACM Transactions on Architecture and Code Optimization (TACO)
- ACM Transactions on Parallel Computing (TOPC)
- Elsevier Future Generation Computer Systems (FGCS)

Invited Talks

- KAIST, POSTECH (Sep 2017)
DeSC: Decoupling Data Supply from Computation for Latency-Tolerant Communication
- DARPA HIVE PI Meeting (Oct 2017)
Graphicionado: A High-Performance and Energy-Efficient Accelerator for Graph Analytics
- HiPEAC 2020 (Jan 2020)
Efficient Data Supply for Parallel Heterogeneous Architectures
- Seoul National University AI Summer School (Aug 2020)
Accelerating Neural Network Attention Mechanism with HW/SW Codesign
- POSTECH Summer AI Seminar (Aug 2020)
Hardware/Software Co-design for Modern AI and Data Analytics Applications

Research Mentoring

During my tenure as a postdoctoral researcher at Seoul National University, I closely mentored and collaborated with the following students (in partnership with Prof. Jae W. Lee), overseeing research projects from initial concept

through system implementation to high-impact publication.

Graduate Students

- **Jaeyoung Jang** (Ph.D., SKKU → Samsung Electronics)
- **Young H. Oh** (Ph.D., SKKU → Samsung Electronics → Ajou University)
- **Jun Heo** (Ph.D., SNU → Samsung Electronics → MangoBoost)
- **Jonghyun Bae** (Ph.D., SNU → LBNL → Google Visiting Researcher → HyperAccel)
- **Shine Kim** (Ph.D., SNU → Samsung Electronics)
- **Seung Yul Lee** (Ph.D., SNU → SNU Postdoc)
- **Yeonhong Park** (Ph.D., SNU → Meta)
- **Wenjing Jin** (Ph.D. student, SNU → Samsung Electronics)
- **Sung Jun Jung** (Ph.D., SNU → SK Hynix)
- **Yejin Lee** (Ph.D., SNU → Meta)
- **Seong Hoon Seo** (Ph.D., SNU)
- **Soosung Kim** (Ph.D., SNU)
- **Jeonghun Gong** (M.S., SNU → Samsung Electronics)
- **Yunho Jin** (M.S., SNU → Harvard Ph.D. → Google)
- **Sam Son** (M.S., SNU → UC Berkeley Ph.D. → Meta)
- **Hyunji Choi** (M.S., SNU → Meta)

Undergraduate Students

- **Jaeyeon Won** (B.S., SNU → MIT Ph.D.)
- **Stuart Sul** (B.S., SNU → Blux → Stanford M.S. → Cursor)
- **U Gyeong Song** (B.S., SNU)

Technical Skills

- **Datacenter Performance & Efficiency:** Fleet-wide resource optimization, performance engineering for distributed systems, capacity efficiency, and infrastructure-scale bottleneck analysis.
- **Hardware/Software Co-Design:** Cross-stack optimization bridging distributed systems and silicon microarchitecture; evaluation, optimization, and architectural guidance for future SoC and server platforms.
- **Distributed Database Systems:** Architecture and optimization of large-scale distributed databases and storage systems; expertise in foundational structures including LSM-trees and Key-Value stores.
- **ML for Systems & Systems for ML:** Leveraging machine learning for system performance, efficiency, and co-design, alongside infrastructure-scale performance optimization for machine learning workloads.
- **Modeling, Simulation & Profiling:** Performance modeling, full-stack system simulation, and architectural evaluation frameworks; **Google-scale production profiling infrastructure** and custom system-level analysis tools.
- **Technical Leadership & Management:** Former Tech Lead Manager (TLM) for a 20-person team at Google; expertise in technical strategy, organizational roadmapping, and research mentorship (Ph.D. level).
- **Implementation:** High-performance C/C++, Python, **LLVM**, and large-scale infrastructure codebase management.